Midterm 2019, "The Mathematical Foundations of Political Methodology"

You may use a calculator and refer to your formula sheet. Please do not use the internet during the exam. Show your work.

1. Consider the following sets of real numbers:

$$\begin{aligned} A &= \{x : 1 \leq x \leq 5\} \\ B &= \{x : 0 \leq x \leq 2\} \\ C &= \{x : x < 0\} \end{aligned}$$

   Describe each of the following sets of real numbers in a single sentence.

   (a) A^c

   (b) $A \cup B$

   (c) $B \cap C^c$

   (d) $(A \cup B) \cap C$

   (e) $(A \cup B)^c \cap C$

2. Find the limit of each of the following sequences as $n \to \infty$, if the limit exists.

   (a) $\frac{5n}{(n^2-5)}$

   (b) $\frac{4n^{\frac{1}{2}}}{(n^2-5)} + 2$

   (c) $\frac{4n}{(n^{\frac{1}{2}}-4)} + 2$

3. Use the (δ, ϵ) definition of a limit to prove that $\lim_{x \to 3} 4x - 11 = 1$

4. Let

$$f(x) = \frac{x}{1 + e^x}$$

   (a) Is f(x) concave, convex or both?

   (b) Find any asymptotes that f may have.

5. Perform the following matrix operations:

   (a) Find the inverse of $\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix}$

(b) Find the rank of $A = \begin{bmatrix} 3 & 1 & 1 & 2 \\ 2 & 1 & -1 & 2 \\ 0 & 3 & 2 & 0 \\ 1 & 0 & 3 & 0 \end{bmatrix}$

(c) What x would make the following matrix singular?

$$\begin{bmatrix} 1 & 7 \\ x & 6 \end{bmatrix}$$

6. If A and B are matricies, which of the following imply the others:

   (a) AB=BA

   (b) Either A or B is the identity matrix.

   (c) B is the inverse of A.

   (d) $A = \frac{1}{det(B)} adj(B)$

   (e) A and B are symmetric.

   (f) A and B are square matrixes

7. What constant k solves the following equation?

$$\int_0^{10} ky^2 dy = 1$$

8. What values of the constant a satisfies the following equation?

$$\int_0^{a} \frac{2x}{2+x^2} dx = 1$$